- The 15-Factor methodology identifies development principles for building applications that offer maximum portability across execution environments, are suitable to be deployed on cloud platforms, can be scaled, guarantee environment parity between development and production, and enable continuous delivery.
- Spring is a suite of projects that provide all of the most common functionality for building modern applications in Java.
- Spring Framework provides an application context in which beans and properties are managed throughout the entire life cycle.
- Spring Boot lays the foundation for cloud native development by speeding up the building of production-ready applications, including embedded servers, auto-configuration, monitoring, and containerization features.
- Container images are lightweight executable packages that include everything needed to run the applications inside.
- Docker is an OCI-compliant platform for building and running containers.
- A Spring Boot application can be packaged as a container image with Cloud Native Buildpacks, a CNCF project that specifies how to convert application source code into production-ready container images.
- When dealing with several containers, as is usually the case in a cloud native system, you need to manage this complex system. Kubernetes provides functionality to orchestrate, schedule, and manage containers.
- Kubernetes Pods are the minimum deployment units.
- Kubernetes Deployments describe how to create application instances as Pods, starting from container images.